

County Wise Analysis of COPD, Heart Disease, Obesity, and Income

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Background

This research performs a county wise comparative analysis of chronic obstructive pulmonary disease (COPD), heart disease, and obesity and their relationships with income.

Also, this research investigates where these issues are the most prominent. Doing so will allow for focused and localized care.



Figure 1:
Star of Life

Research Questions

- 1) What are the relationships between median household income and the prevalence of COPD, heart disease, and obesity for counties in the United States
- 2) What areas of the United States are the poorest and have the highest prevalence of the disease that poses the biggest issue within the United States?

Related Research

- "Poverty, a risk factor overlooked: a cross-sectional cohort study comparing poverty rate and cardiovascular disease outcomes in the state of Florida"
- "Increasing Trajectories of Multimorbidity Over Time: Birth Cohort Differences and the Role of Changes in Obesity and Income"

Data

1. Title: "PLACES: Local Data for Better Health, County Data 2021 release"

Description: It describes the prevalence of many diseases, such as COPD, heart disease, and obesity, for counties in the United States in 2019. Also, it is a 44 MB .csv file with 188,000 rows.

Source: Published by the Centers for Disease Control and Prevention (CDC) and last updated in 2022

2. Title: "INCOME IN THE PAST 12 MONTHS (IN 2021 INFLATION-ADJUSTED DOLLARS)"

Description: It consists of income values collected from 3220 counties in the United States in 2019 that are adjusted for 2021 inflation dollars. Also, it is a 3.68 MB .csv file.

Source: Published by the United States Census Bureau and last updated in 2021

Preprocessing

Steps needed to process the data into a format for ggplot2 and Tableau:

1. Import the datasets from the CDC and United States Census Bureau's databases into RStudio
2. Remove data for U.S. territories for both datasets
3. Filter for prevalence of just COPD, heart disease, and obesity for each county in the "PLACES: Local Data for Better Health, County Data 2021 release" dataset
4. Separate county names from state names that were previously stored in the same column for the "INCOME IN THE PAST 12 MONTHS (IN 2021 INFLATION-ADJUSTED DOLLARS)" dataset
5. Combine both datasets into one dataset where each row describes 2019 median household income, COPD, heart disease, and obesity prevalence for a county in the U.S.

	State		Median	COPD	Heart Disease	Obesity
Year	Abbr.	County	Income (\$)	(%)	(%)	(%)
2019	LA	East Carroll	22,346	13.2	11	47.3
2019	GA	Clay	22,325	14.7	13.2	44.3
2019	MS	Holmes	21,504	11.4	9.8	48.6

Table 1: Top 3 lowest income counties in the U.S.

	State		Median	COPD	Heart Disease	Obesity
Year	Abbr.	County	Income (\$)	(%)	(%)	(%)
2019	VA	Loudoun	142,299	3.9	3.8	26.6
2019	VA	Fairfax	124,831	4.3	4.4	24.6
2019	CA	Santa Clara	124,055	3.5	3.9	21.4

Table 2: Top 3 highest income counties in the U.S.

Methods

- Used ggplot2 to create a facet plot to perform comparative analysis of the relationships between median household income and prevalence of COPD, heart disease, and obesity.
- Used Tableau to create choropleth maps to understand the areas of the United States that are the poorest and have the highest prevalence of the disease that is the most prominent issue.

Results

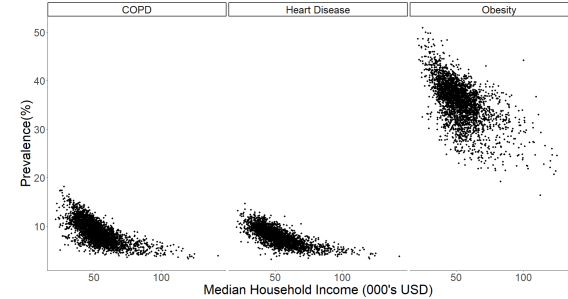


Figure 2: Prevalence(%) of COPD, heart disease, and obesity vs. median household income (000's USD) for counties across the U.S.

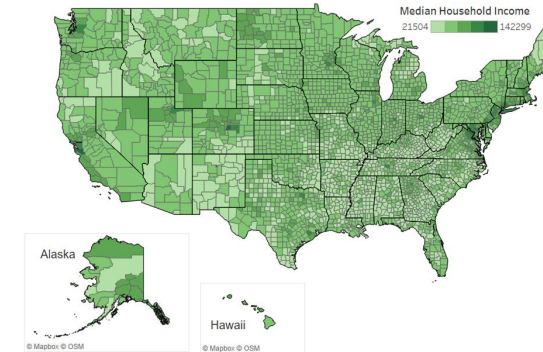


Figure 3: Median household income for counties across the U.S.

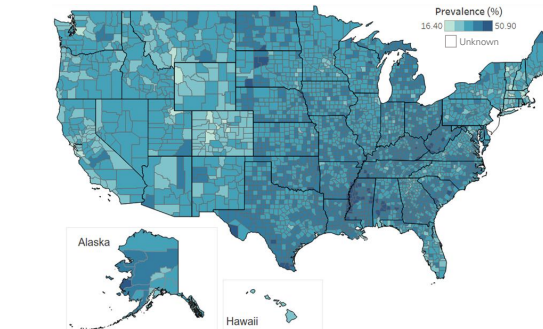


Figure 4: Prevalence (%) of obesity for counties across the U.S.

Discussion

The prevalence of COPD, heart disease, and obesity have negative associations with median household income, as shown in Fig. 2. However, obesity is by far the most prevalent disease in most U.S. counties compared to COPD and heart disease. Also, most counties in the Southeast region of the U.S., New Mexico, Maine, and Idaho have low median household income and a moderate to high prevalence of obesity, as depicted in Fig. 2 and Fig. 3. In all, obesity poses the largest issue for counties in the U.S. compared to COPD and heart disease. Also, more community aid services need to be present to aid counties in the Southeast region, New Mexico, Maine, and Idaho because they are hit the hardest with obesity and low income. However, the disease dataset did not contain New Jersey obesity data.

Conclusion

It is not only important to know the relationships between certain diseases and income. It is also important to understand where these health and financial issues are the most prominent. That way, policy makers and groups can make decisions on where to send aid or do more to reduce hardships.

Future Work

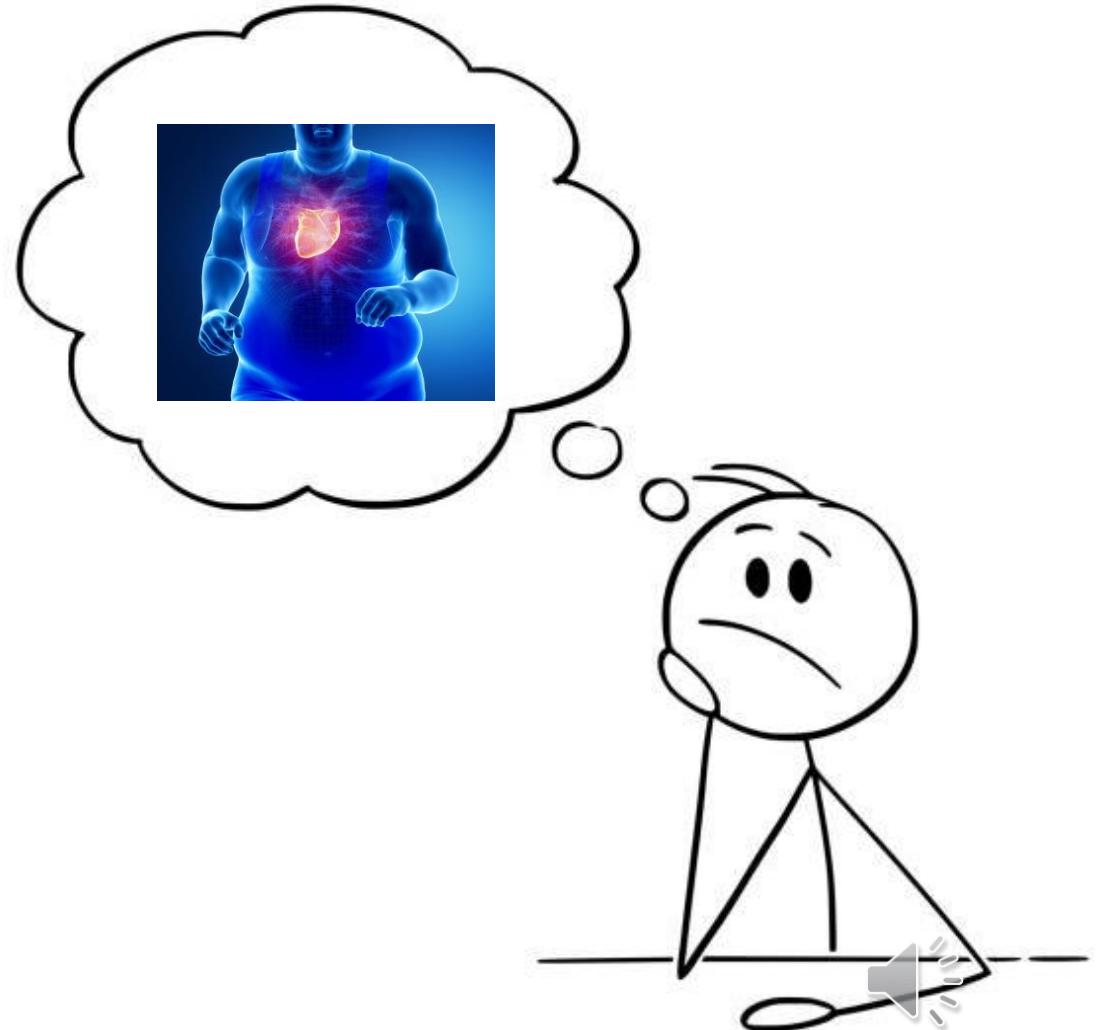
- I. Acquire more recent and complete income and obesity data for counties in every U.S. state
- II. Explore more diseases, such as cancer and mental health
- III. Compare the U.S. to other countries, such as Canada

References

- Al-Turk, B., Harris, C., Nelson, G., Smotherman, C., Palacio, C., & House, J. (2018). Poverty, a risk factor overlooked: a cross-sectional cohort study comparing poverty rate and cardiovascular disease outcomes in the state of Florida. *Journal of Investigative Medicine*, 66(3), 693-695.
- Canizares, Hogg-Johnson, S., Gignac, M. A. M., Glazier, R. H., & Badley, E. M. (2018). Increasing Trajectories of Multimorbidity Over Time: Birth Cohort Differences and the Role of Changes in Obesity and Income. *The Journals of Gerontology, Series B, Psychological Sciences and Social Sciences*, 73(7), 1303-1314.
- Center for Disease Control and Prevention. (2022). PLACES: Local Data for Better Health, County Data 2021 release [Data set]. <https://chronicdata.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-2019we5-unb>
- Emergency Medical Services. (2006). *The Star of Life* [Image].
- United States Census Bureau. (2021). *Income in the past 12 months (in 2020 inflation-adjusted dollars)* [Data set]. <https://data.census.gov/cedsci/table?q=income%20by%20county&tid=ACSS5Y2020.S1901>

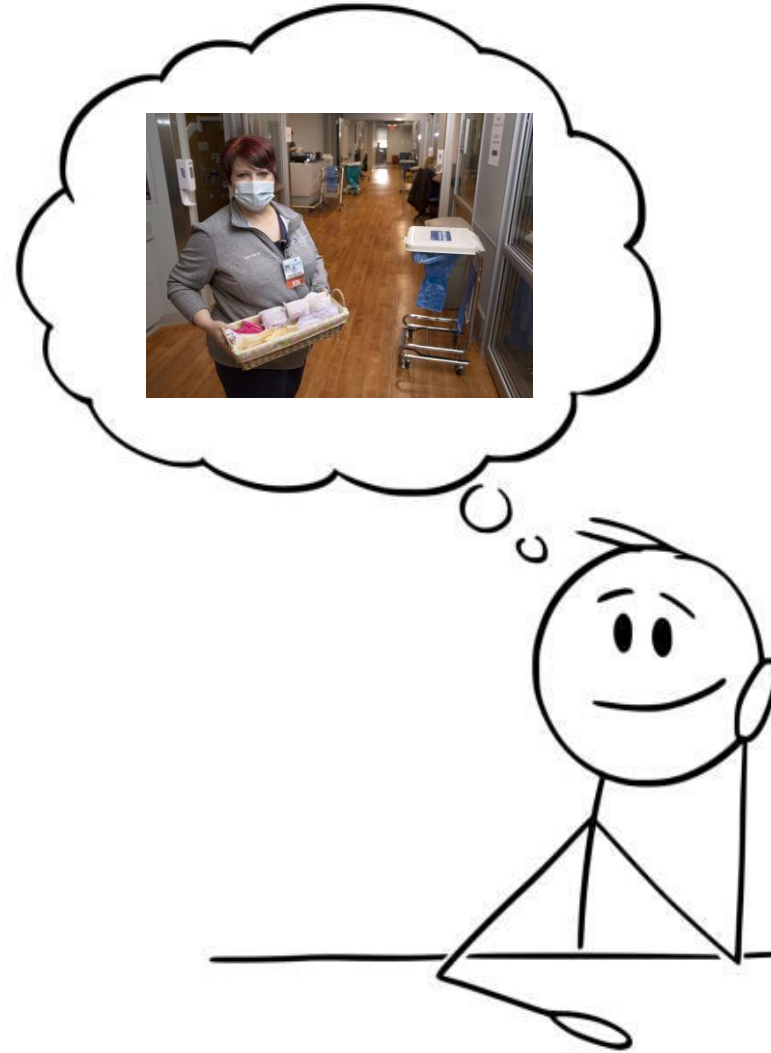
What are we talking about?

- Perform a county wise comparative analysis of the relationships between median household income and the prevalence of chronic obstructive pulmonary disease (COPD), heart disease, and obesity
- Investigate where these financial and health issues are the most prominent



Why does it matter?

- Prioritize issues
- Understand where aid needs to be sent or improved policies should be created.



<https://pennstatehealthnews.org/2021/03/go-with-the-flo-nurse-spearheads-project-to-combat-period-poverty/>

<https://www.istockphoto.com/photos/stick-figure-with-thought-bubble>

Related Research

1. Title: “Poverty, a risk factor overlooked: a cross-sectional cohort study comparing poverty rate and cardiovascular disease outcomes in the state of Florida”

Findings: In Florida, as poverty rates increase, so do cardiovascular disease, mortality rates, and hospitalizations

2. Title: “Increasing Trajectories of Multimorbidity Over Time: Birth Cohort Differences and the Role of Changes in Obesity and Income”

Findings: Larger percentage of lower income individuals tended to have multimorbidity compared to wealthier income individuals according to a voluntary sample of Canadians



Research Questions

RQ1: What are the relationships between median household income and the prevalence of COPD, heart disease, and obesity for counties in the United States?

RQ2: What areas of the United States are the poorest and have the highest prevalence of the disease that poses the biggest issue within the United States?



Datasets Used for Analysis

Dataset 1:

Title: “PLACES: Local Data for Better Health, County Data 2021 release”

Description: Contains data pertaining to the prevalence of many diseases, such as COPD, heart disease, and obesity for counties in the United States in 2019.

Source: Published by the Centers for Disease Control and Prevention (CDC) and last updated in 2022 from <https://chronicdata.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-20/swc5-untb>

Dataset 2:

Title: “INCOME IN THE PAST 12 MONTHS (IN 2021 INFLATION-ADJUSTED DOLLARS)”

Description: Contains data pertaining to income values collected from counties in the United States in 2019 that are adjusted for 2021 inflation dollars.

Source: Published by the United States Census Bureau and last updated in 2021 from <https://data.census.gov/cedsci/table?q=income%20by%20county&tid=ACST5Y2020.S1901>



Data Preprocessing

1. Import the datasets into RStudio
2. Remove data for U.S. territories
3. Filter for prevalence of just COPD, heart disease, and obesity for each county
4. Separate county names from state names that were previously stored in the same column
5. Combine both datasets into one dataset where each row describes 2019 median household income, COPD, heart disease, and obesity prevalence for a county in the U.S.

Year	State Abbr.	County	Median Income (\$)	COPD (%)	Heart Disease (%)	Obesity (%)
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2019	CA	Santa Clara	124,055	3.5	3.9	21.4

Table 2: Top 3 highest income counties in the U.S.

Methods

- Used ggplot2 to create a facet plot showing the relationships between median household income and prevalence of COPD, heart disease, and obesity
- Used Tableau to create choropleth maps describing median household income and prevalence of the disease that poses the most prominent issue for counties in the U.S.



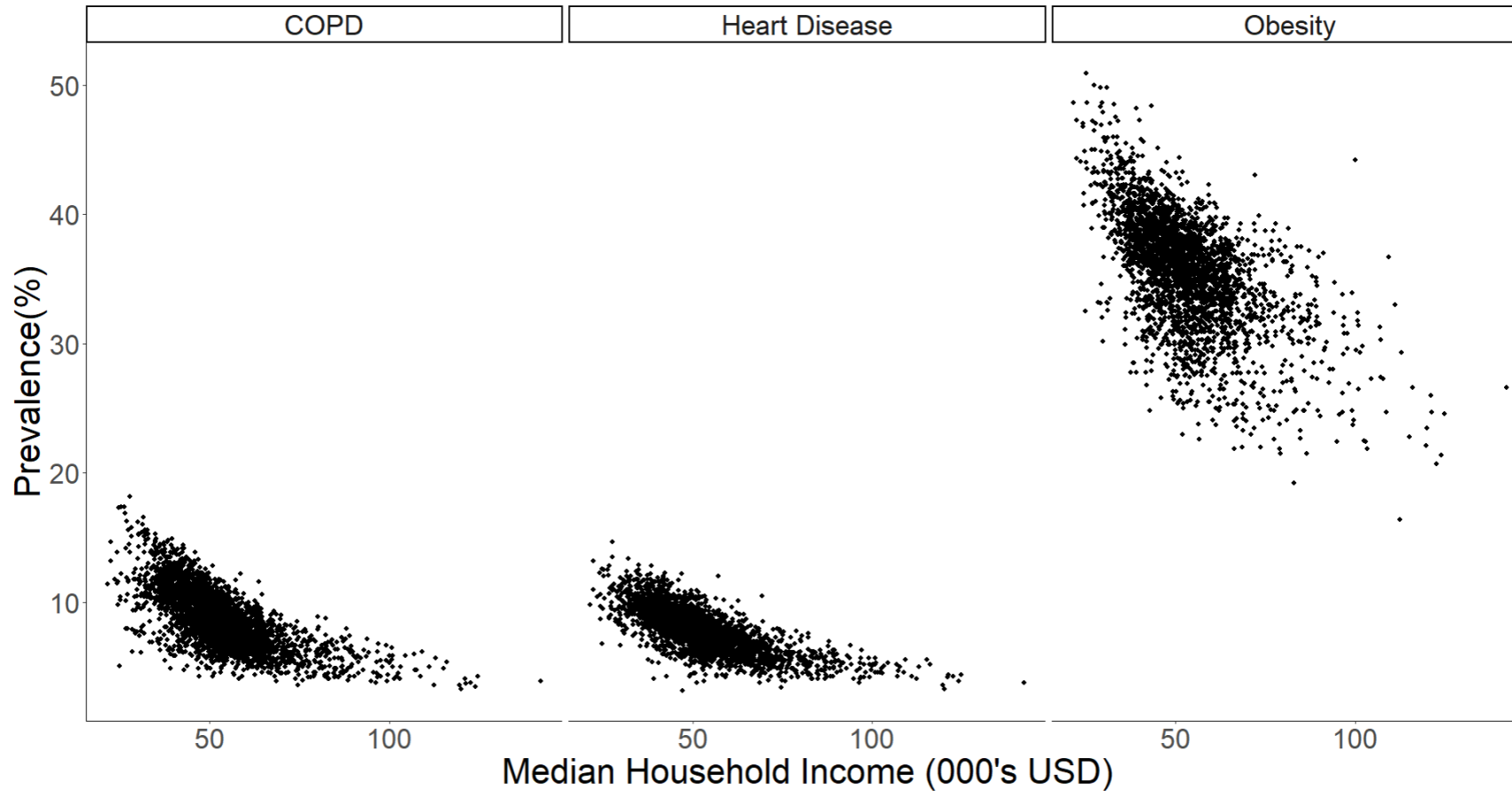
<https://ggplot2.tidyverse.org/>



<https://logos-world.net/tableau-logo/>



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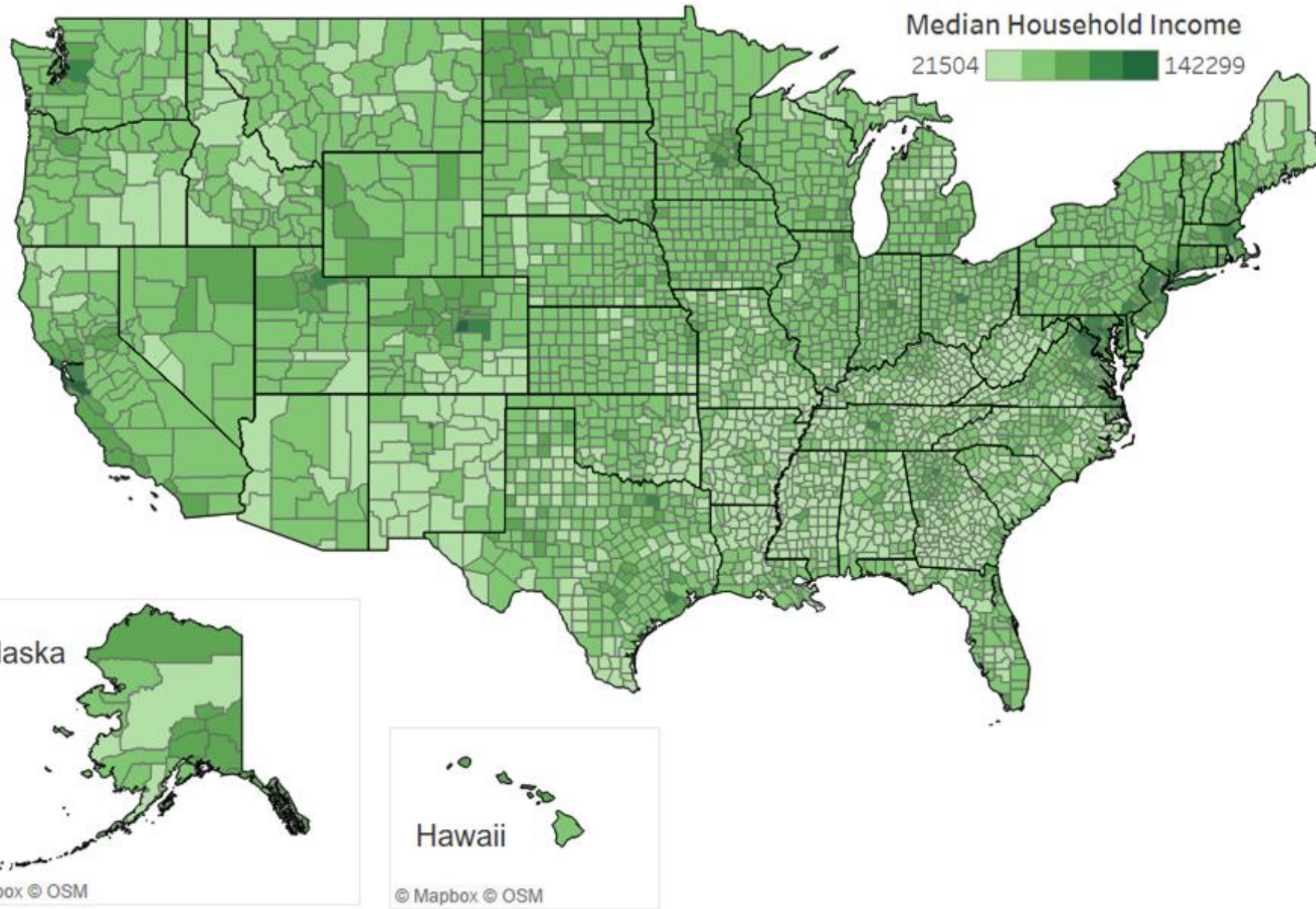


Key findings:

- Negative associations with median household income.
- Obesity is the most prominent issue



Median household income for counties across the U.S.

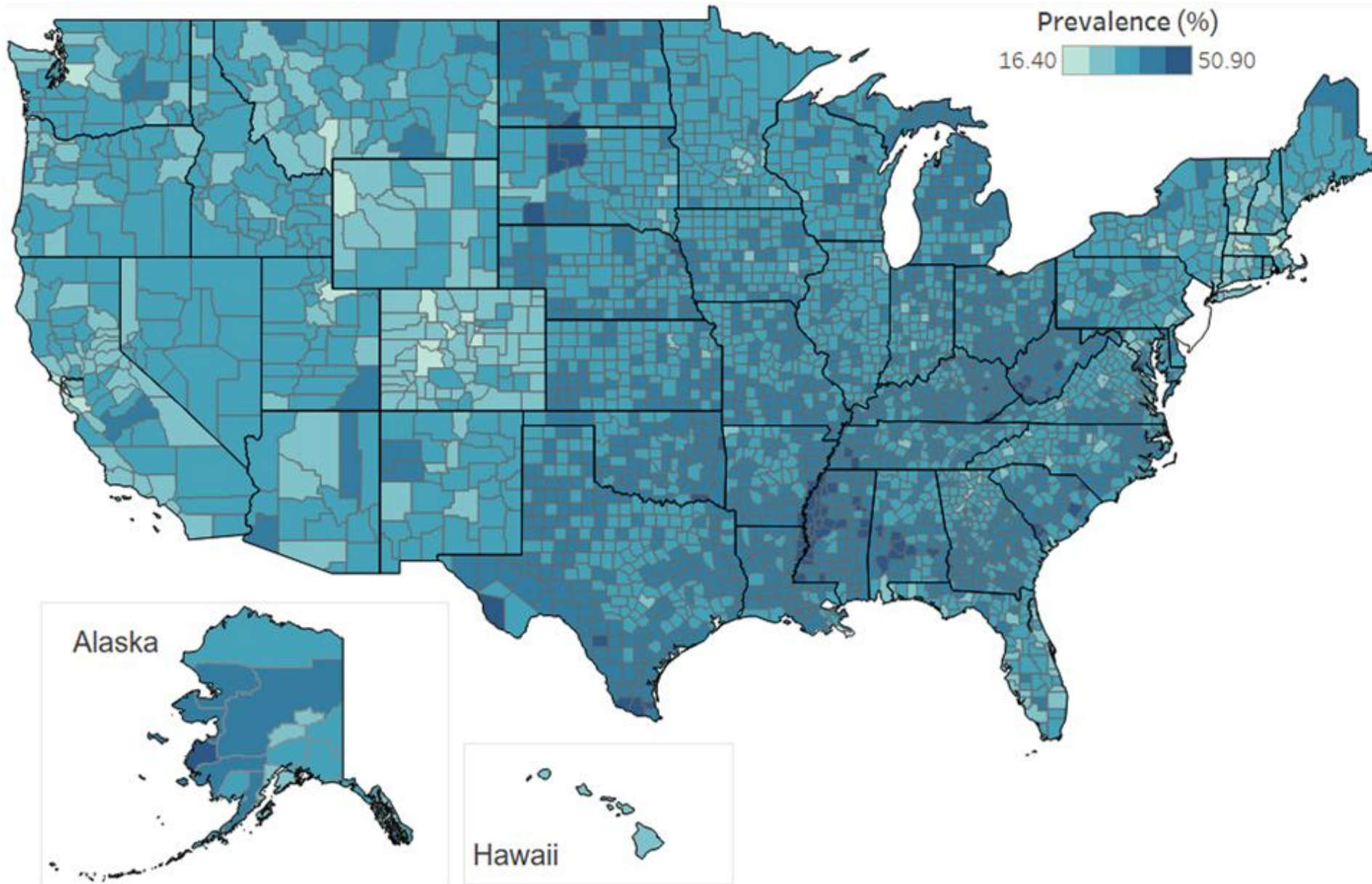


Key findings:

- Most counties in the Southeast region, New Mexico, Maine, and Idaho have low median household incomes.



Prevalence (%) of obesity for counties across the U.S.



Key findings:

- Most counties in the Southeast region, New Mexico, Maine, and Idaho have a moderate to high prevalence of obesity.
- Missing New Jersey obesity data



Discussion

RQ1: What are the relationships between median household income and the prevalence of COPD, heart disease, and obesity for counties in the United States?

Answer: The prevalence of COPD, heart disease, and obesity have negative associations with median household income. However, obesity is the most prevalent disease in most U.S. counties compared to COPD and heart disease.

RQ2: What areas of the United States are the poorest and have the highest prevalence of the disease that poses the biggest issue within the United States?

Answer: Most counties in the Southeast region, New Mexico, Maine, and Idaho have low median household income and a moderate to high prevalence of obesity



Future Research

- Acquire more recent and complete income and obesity data for counties in every U.S. state
- Explore more diseases, such as cancer and mental health
- Compare the U.S. to other countries, such as Canada



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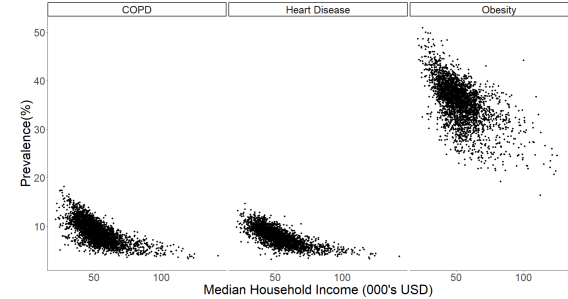


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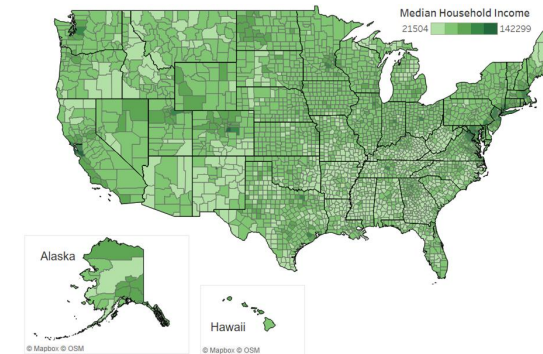


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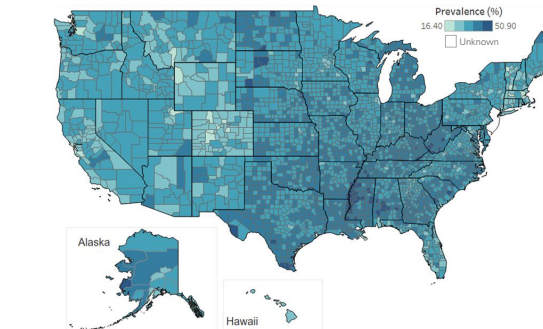


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